

Pre-work · Intro to Anatomy

Intro to Anatomy. Read the Part A chapters first, then work through this in order. When you walk into class you should be ready to use this material, not hearing it for the first time.

Module 1 · 1 of 4

Bring this to class

How to work this pre-work

1 min

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

- 1 Start with the reading.**
Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.
- 2 Organize what you read.**
In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.
- 3 Draw it.**
In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.
- 4 Apply it.**
Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.
- 5 Come back to it later in the week**
, not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

Organize it

2 min

PANEL 1 OF 3

Turn the two intro chapters into mini visuals. Seven chunks, each in the shape that fits it.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1 LADDER

The six levels of structural organization

Six rungs, simplest at the bottom. Write one real example beside each rung, not the one from the notes.

2 SPLIT

The seven directional pairs

One split per pair. Pair on the branches, and underneath, the one sentence that proves you can use it. Star the two you are least sure of.

3 GRID

Planes crossed with what they divide

Planes down the side, the two parts each produces across the top. Six rows. Draw the cut line on a stick figure in one box.

4 HUB

Body cavities

Ventral and dorsal as two hubs. Spoke out to each subdivision, then to what it contains. Mark the diaphragm on the spoke it actually sits on.

5 CHAIN

Serous membrane, wall to organ

Body wall, arrow, parietal layer, arrow, cavity with fluid, arrow, visceral layer, arrow, organ. Then write the chain three times, once each for pericardium, pleura, peritoneum.

6 HUB

The mediastinum

Borders as spokes on one side, contents as spokes on the other. Five borders, five contents.

7 GRID

Quadrants and regions on one abdomen

Draw the abdomen once. Quadrant lines in one color, the nine-region lines in a second. Label the organs where they sit. One drawing does both systems.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

Draw it

2 min

PANEL 2 OF 3

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

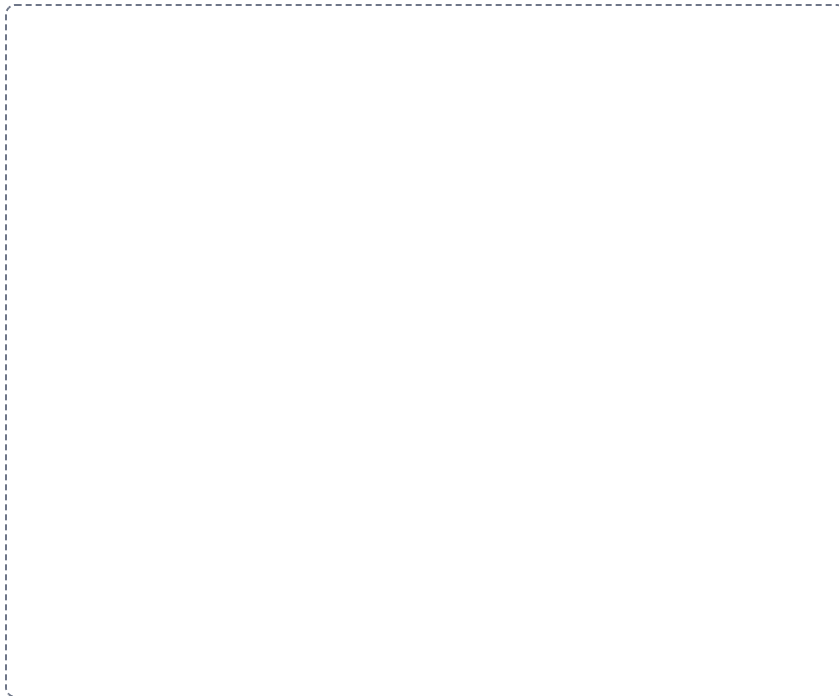
DRAWING 1

Anatomical position, with the planes on it

BRAIN DUMP FIRST

Two minutes. Anatomical position and every plane you can name, with what each one separates.

Draw a full figure in anatomical position. Mark the five features that define it: feet flat, arms at the sides, palms forward, head and eyes forward, body erect. Then cut the same figure with the sagittal, frontal, and transverse planes. Label each, and write the two portions each one creates.



In your notebook, head the page **M1-1 Anatomical position, with the planes on it** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can use midsagittal and parasagittal on one figure and say the difference in a sentence.

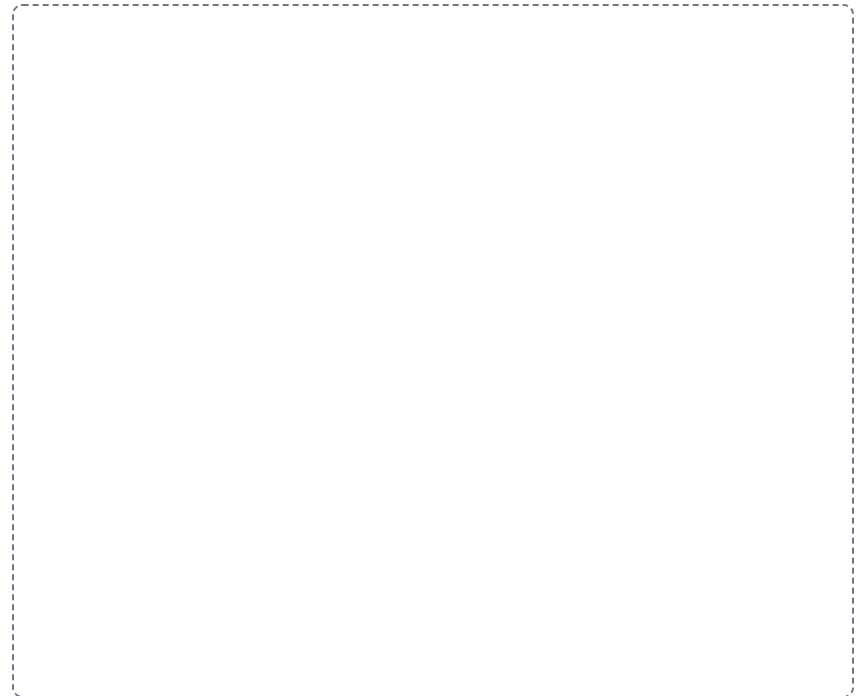
DRAWING 2

The cavities, lateral view

BRAIN DUMP FIRST

Two minutes. Every cavity, what is in it, and what separates it from the one next door.

Draw the body in profile. Outline the dorsal cavity and the ventral cavity, then subdivide each. Shade the thoracic and abdominopelvic cavities differently and draw the diaphragm between them. Put the mediastinum in, and shade the retroperitoneal space behind the peritoneum.



In your notebook, head the page **M1-2 The cavities, lateral view** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can point at any organ on your drawing and say which cavity it sits in and which membrane wraps it.

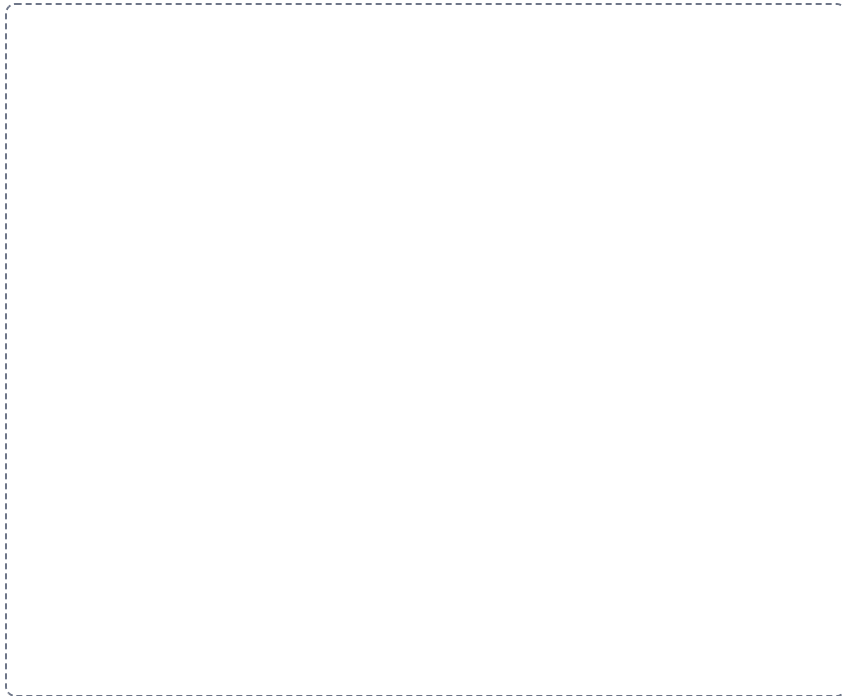
DRAWING 3

Serous membrane, the fist and the balloon

BRAIN DUMP FIRST

One minute. Parietal, visceral, cavity, fluid. What each one touches.

Push a fist into a partly inflated balloon and draw that. The layer against your fist is visceral, the outer layer is parietal, the space between holds the fluid. Label all four. Then draw what happens to the space when it fills with air, and when it fills with fluid.



In your notebook, head the page **M1-3 Serous membrane, the fist and the balloon** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can explain a collapsed lung and a friction rub using only this drawing.

Apply it

2 min

PANEL 3 OF 3

First pass with everything closed. Then open the notes in a second color and mark what you missed.

WORKED EXAMPLE

The elbow is _____ to the wrist.

Answer. Proximal.

Reasoning. Both structures are on the same limb, so the proximal and distal pair is available. Proximal means nearer the point of attachment to the trunk. The elbow attaches nearer the shoulder than the wrist does, so the elbow is proximal. Note the wrong answer you could reach for: the elbow is also superior to the wrist in anatomical position, but that is the weaker answer, because the question is about a limb and the limb-specific pair is the precise one.

Set A · Directional relationships

1. The sternum is _____ to the vertebral column.
2. The lungs are _____ to the diaphragm.
3. The thumb is _____ to the little finger, in anatomical position.
4. The skin of the thigh is _____ to the femur.
5. A stroke in the left cerebral hemisphere causes weakness on the right side of the body. That effect is _____.
6. Explain in one sentence why it is wrong to say the nose is proximal to the chin.

Set B · Planes and sections

1. A scan of the head shows both eyes and both ears in the same image, with the top of the skull missing. Which plane produced it, and how do you know?

2. A cut separates the body into anterior and posterior portions. Name the plane.

3. A section through the arm shows a single round bone surrounded by a ring of muscle. Which plane, and how do you know?

4. You want an image showing the relationship between the front of the heart and the sternum. Which plane serves you best, and why?

Set C · Cavities and membranes

1. A stab wound opens the pleural cavity. Predict what happens to the lung on that side, and say why.

2. A patient has a ruptured appendix. Name the membrane involved, the condition it produces, and why the problem does not stay local.

3. A surgeon plans to reach the right kidney. Is the approach through the peritoneal cavity or behind it? Name the term and one other structure in the same space.

4. A friction rub is heard over the heart. Name the two layers rubbing, the fluid that should be preventing it, and the condition.

Set D · Locate it

1. A patient reports pain in the right lower quadrant. Name two organs that could be responsible.

2. Give the regional term for the front of the elbow, the back of the knee, and the sole of the foot.

3. Sort as axial or appendicular: brachial, thoracic, popliteal, vertebral, carpal.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

Mini visual sheet

2 min

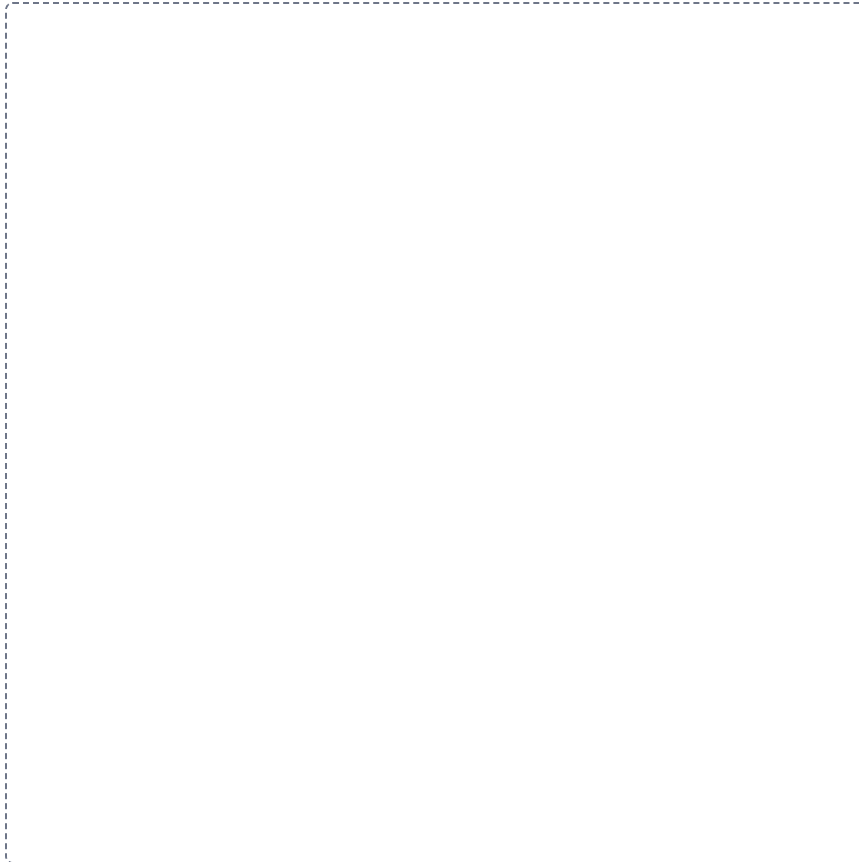
Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

M1-1

The six levels of structural organization

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

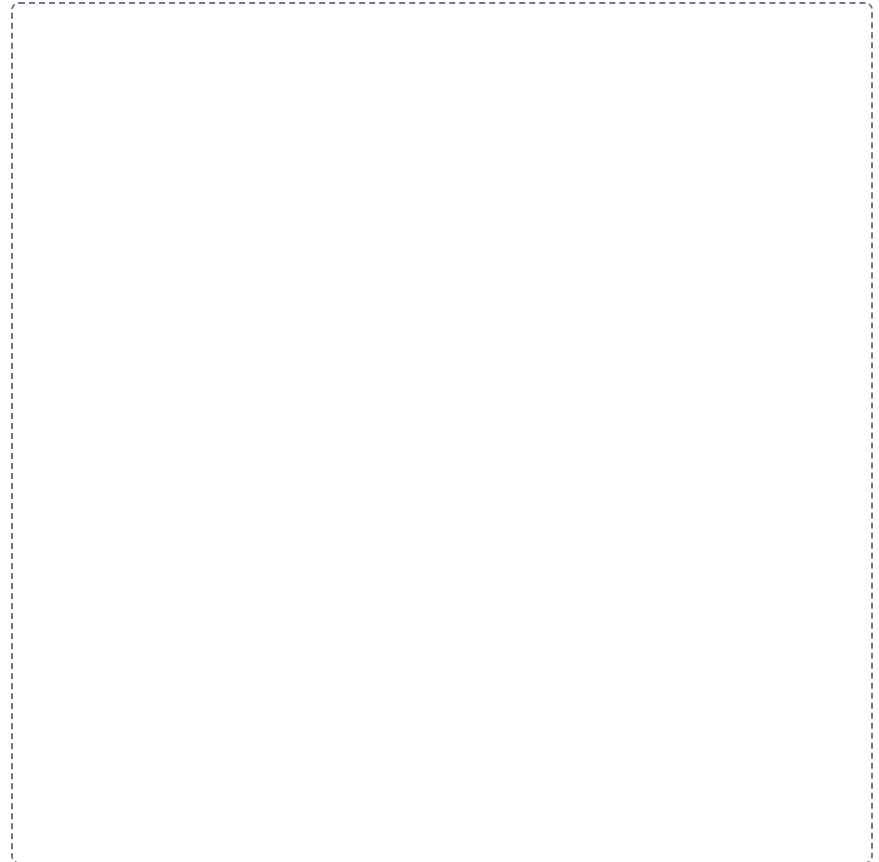


M1-2

The seven directional pairs

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

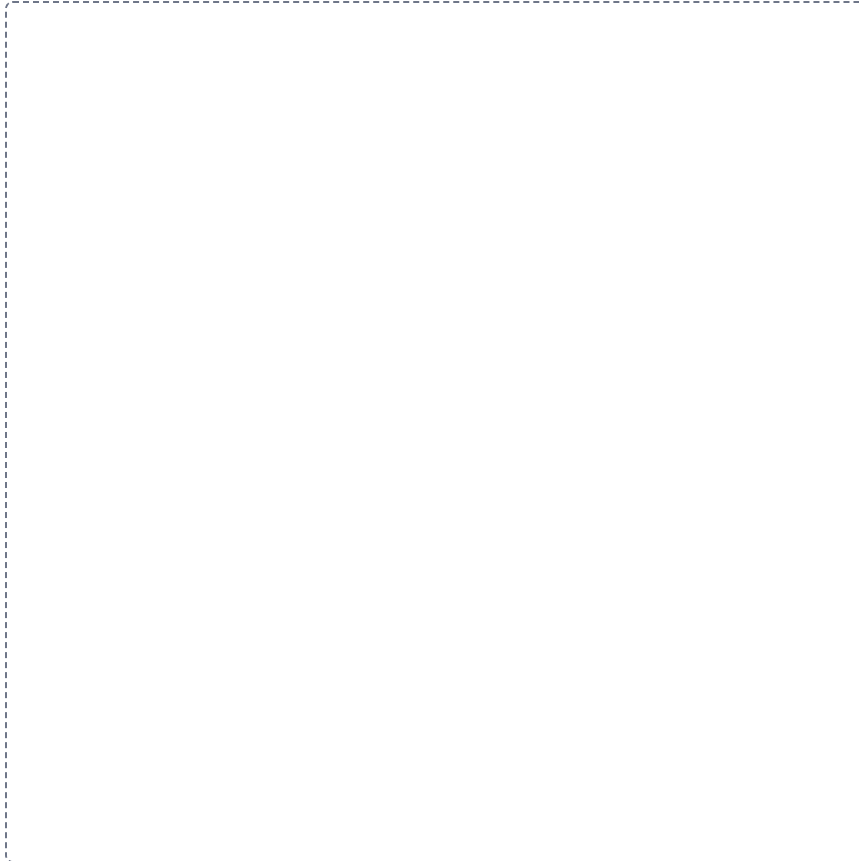


M1-3

Planes crossed with what they divide

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

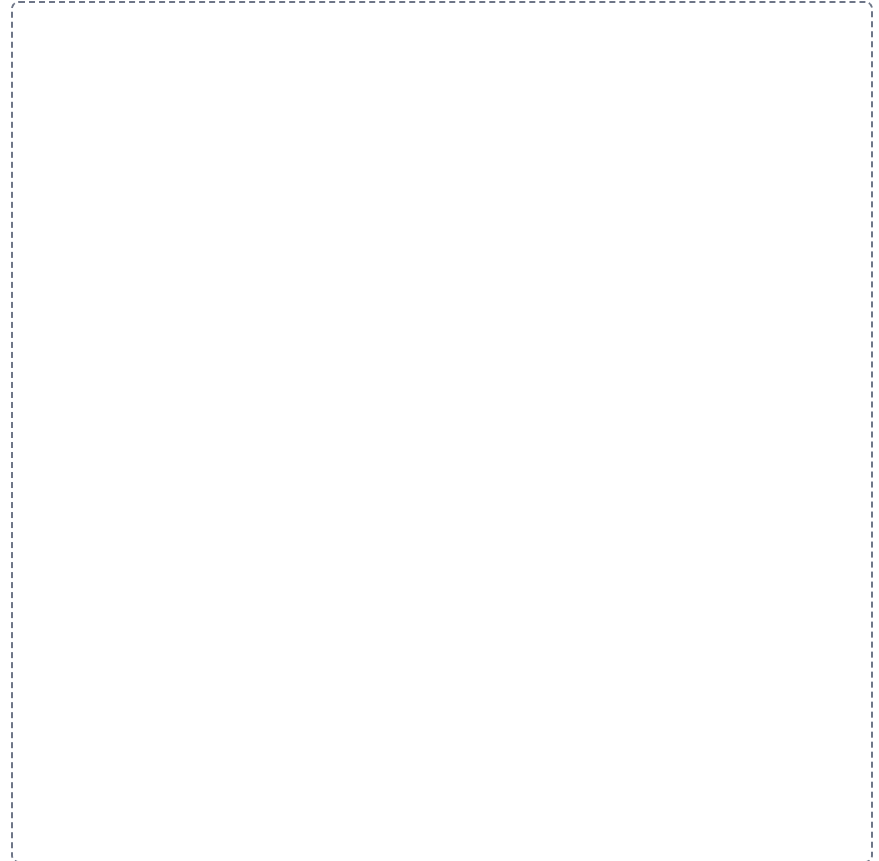


M1-4

Body cavities

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

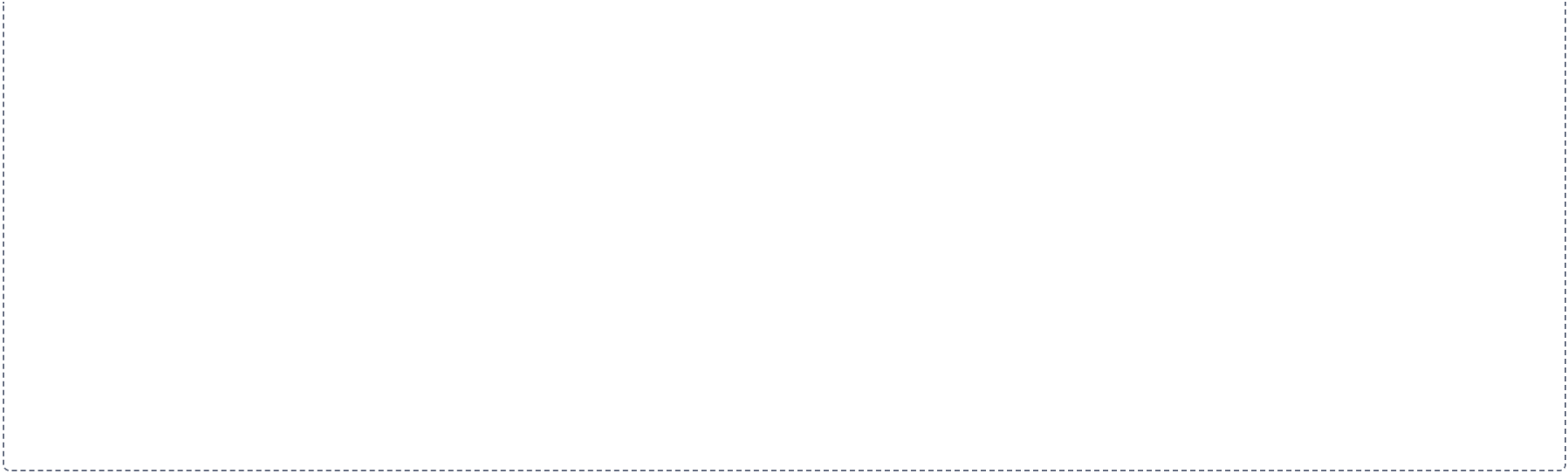


Serous membrane, wall to organ

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

A large dashed rectangular box intended for a student to draw a flowchart. The box is empty and occupies the central portion of the page below the instructions.

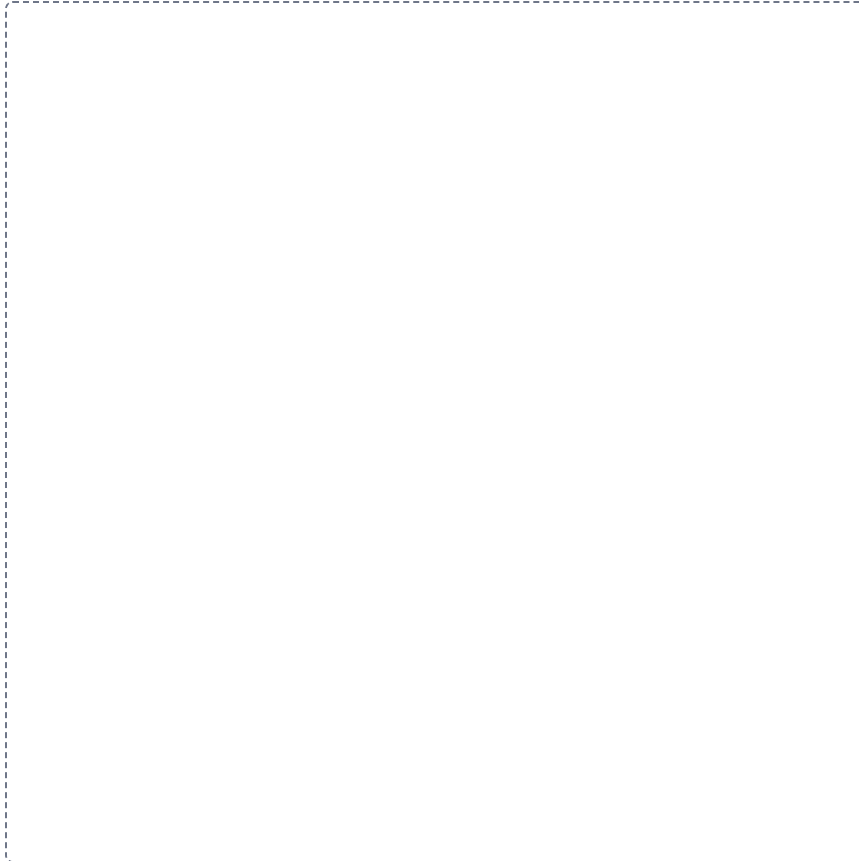


M1-6

The mediastinum

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.



M1-7

Quadrants and regions on one abdomen

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

